WORKED SOLUTION

STEP I 1991, Question 9

Pure • Inequalities and Harmonic Series

STEP I 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q9

The key idea

In part (i), each inequality traps x between two kth roots, and the trick is to find one lower bound and one upper bound that clash. To compare a cube root with a fifth root without a calculator, raise both sides to the fifteenth power. In part (ii) we bound a long sum below by (number of terms) $\times$ (smallest term). Then we chain disjoint blocks of that shape together to make the harmonic series as big as we like.

Solution

Part (i)

Since $x > 1$, every power of x is positive, and raising both sides of an inequality between positive numbers to a positive power keeps its direction. We use this throughout.

Suppose, for a contradiction, that $n \geq 5$. Then in particular the third and fifth inequalities hold:

$$x^3 > 3 \qquad \text{and} \qquad x^5 < 6.$$

Raise the first to the fifth power and the second to the cube:

$$x^{15} > 3^5 = 243 \qquad \text{and} \qquad x^{15} < 6^3 = 216.$$

These give $243 < x^{15} < 216$, which is impossible. So $n \geq 5$ can't happen, and therefore $n \leq 4$.

? **Ask yourself** How would you know to pick the third and fifth inequalities?

Each inequality says $k^{1/k} < x < (k+1)^{1/k}$. The lower bounds $k^{1/k}$ go 1, 1.41, 1.44, 1.41, 1.38, …, largest at $k = 3$. The upper bounds $(k+1)^{1/k}$ decrease, and $6^{1/5} \approx 1.43$ is the first to drop below $3^{1/3} \approx 1.44$. A rough mental estimate points to the pair, and the fifteenth powers then prove it exactly. (The value $n = 4$ really can occur. For example $x = 1.45$ gives $x^2 = 2.1025$, $x^3 \approx 3.05$, $x^4 \approx 4.42$.)

Part (ii)

Let $n \geq 2$. The sum $\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n^2}$ has $n^2 - n$ terms, and each is at least $\frac{1}{n^2}$, the last one. Since $n \geq 2$ there are $n^2 - n \geq 2$ terms, so the first term $\frac{1}{n+1}$ is strictly bigger than $\frac{1}{n^2}$. Hence

$$\frac{1}{n+1} + \cdots + \frac{1}{n^2} > (n^2 - n) \cdot \frac{1}{n^2} = 1 - \frac{1}{n}.$$

Adding $\frac{1}{n}$ to both sides,

$$\frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{n^2} > 1,$$

as required.

Making the harmonic sum exceed 10 Each block $\frac{1}{m} + \cdots + \frac{1}{m^2}$ with $m \geq 2$ contributes more than 1. To use several of them we need blocks that don't overlap, so we start each new block just after the last one ends. Define

$$a_1 = 2, \qquad a_{k+1} = a_k^2 + 1,$$

so $a_1 = 2$, $a_2 = 5$, $a_3 = 26$, $a_4 = 677, \ldots$. Block k runs from $\frac{1}{a_k}$ to $\frac{1}{a_k^2}$, and the next block starts at $\frac{1}{a_k^2+1} = \frac{1}{a_{k+1}}$. So the blocks for $k = 1, \ldots, 9$ are disjoint and consecutive, and together they cover $\frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{a_9^2}$. Therefore

$$\sum_{n=1}^{a_9^2} \frac{1}{n} = 1 + \sum_{k=1}^{9} \left(\frac{1}{a_k} + \cdots + \frac{1}{a_k^2} \right) > 1 + 9 = 10.$$

Ask yourself a_9^2 is a gigantic number. Can we name an explicit N instead?

Yes, by bounding the a_k. We claim $a_k < 2^{2^k}$. This holds for $k = 1$ since $2 < 4$. If $a_k < 2^{2^k}$, then $a_k \leq 2^{2^k} - 1$, so

$$a_{k+1} = a_k^2 + 1 \leq (2^{2^k} - 1)^2 + 1 = 2^{2^{k+1}} - 2^{2^k+1} + 2 < 2^{2^{k+1}}.$$

So by induction $a_9 < 2^{512}$ and $a_9^2 < 2^{1024}$. Adding more positive terms only makes the sum bigger, so

$$N = 2^{1024} \quad \text{gives} \quad \sum_{n=1}^{N} \frac{1}{n} > 10.$$

Any larger N works too.

Otherwise: grouping into blocks $\frac{1}{2^j+1} + \cdots + \frac{1}{2^{j+1}}$, each of 2^j terms that are all at least $\frac{1}{2^{j+1}}$, shows that each block exceeds $\frac{1}{2}$ for $j \geq 1$ and equals $\frac{1}{2}$ for $j = 0$. So $\sum_{n=1}^{2^{18}} \frac{1}{n} > 1 + \frac{18}{2} = 10$, and $N = 2^{18} = 262\,144$ is a much smaller answer.

Checks

- Part (i): numerically, the inequalities leave $x \in (3^{1/3}, 5^{1/4}) \approx (1.442, 1.495)$ when $n = 4$. Adding the fifth inequality forces $x < 6^{1/5} \approx 1.431$, so nothing is left.
- Part (ii): with exact fractions, $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12} > 1$, and the block sum exceeds 1 for every n from 2 to 39.
- The harmonic sum first passes 10 at $N = 12\,367$, so every valid answer must be at least this. Both 2^{18} (where the sum is about 13.05) and 2^{1024} are comfortably bigger.

Common slips

- Raising inequalities to powers without noting that both sides are positive (here guaranteed by $x > 1$).
- Using blocks $[2, 4]$, $[4, 16]$, $[16, 256]$, …, which share their endpoints, so some terms are counted twice.
- Writing $\geq$ instead of $>$ in part (ii) without checking that at least one term is strictly larger than $\frac{1}{n^2}$.

Beyond STEP

Part (ii) is a version of the Cauchy condensation idea. It proves that the harmonic series diverges, though astonishingly slowly: its partial sums grow like $\ln N + \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Comparing a sum with an integral gives this growth rate and makes the smallest N easy to estimate ($e^{10-\gamma} \approx 12\,367$). Abbott's *Understanding Analysis* treats condensation and series convergence carefully.

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